

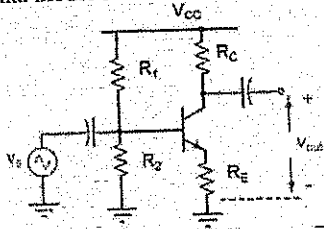
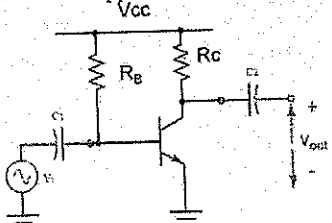
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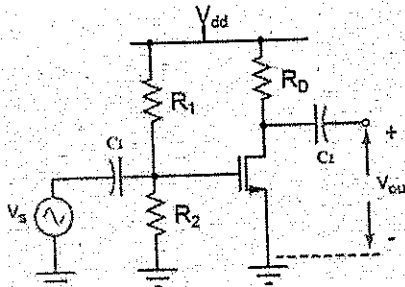
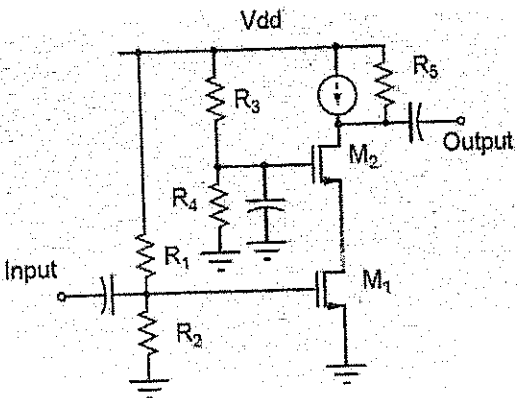
Roll no:

Degree: B. Tech. Semester: 3rd — Course work
END-SEMESTER EXAMINATION, Nov-Dec 2023

M.M: 40 Course Title: Microelectronics Circuits and Applications
Duration: 3 Hours Course Code: EIECC07

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q.no	Question	Marks	CO
Q1	Attempt any two parts of the following		
1(a)	For the circuit shown in Fig.1, determine R_{in} , R_o and A_v with the help of small signal model of transistor.  Fig.1	4	CO1
1(b)	For the circuit shown in Fig.2, given that $V_{CC} = 12V$; $\beta = 100$; $V_{BE(ON)} = 0.6V$; $R_B = 570k\Omega$; $R_C = 3.3k\Omega$; $C_1 = C_2 = 10\mu F$; $C_L = 100pF$. Calculate the following i. Operating points ii. Small signal parameters iii. Voltage gain iv. Lower and upper cutoff frequencies  Fig.2	4	CO1

1(c)	For the circuit given in Fig.3, given that $(K.W/L) = 1mA/V^2$ $V_t = 1V$, $V_{dd} = 12V$, $R_1 = 9k$, $R_2 = 3k$, $R_D = 3k$. $C_1 = C_2 = 10\mu F$ and $C_L = 100pF$. Then calculate R_{in} , R_o , A_v and upper and lower cut-off frequency of amplifier also draw the frequency response of amplifier.  Fig.3	4	CO1
Q2	Attempt any two parts of the following		
2(a)	For the Cascode amplifier shown in Fig.4, determine input impedance, output impedance and voltage gain  Fig.4	4	CO2

2(b)	For the circuit shown in Fig.5, given that $V_{CC} = 12V$; $\beta_1 = \beta_2 = 100$; $V_{BE1(ON)} = V_{BE1(ON)} = 0.6V$; $R_{E1} = R_{E2} = 5.2k\Omega$; $R_f = 1k\Omega$; $V_{A1} = V_{A2} = 100V$ and $V_{INC} = 2.6V$. Calculate differential and common mode gain of the circuit.	4	CO2
	Fig.5		
2(c)	With proper circuit diagram, explain the working of differential amplifier. Why common mode gain of the amplifier should be low?	4	CO2
Q3	Attempt any two parts of the following		
3(a)	What are the advantages of OTA over OP-AMP. For the circuit given in Fig.6 calculate the output voltage V_o and identify the filters, which are realized by the given circuit	4	CO3
	Fig.6		
3(b)	For the circuit given in figure below, determine V_o/V_{in} and output impedance (R_o)	4	CO3

	(a) (b)		
3(c)	Draw the circuit diagram of KHN-filter and determine all the parameters of the realized filters	4	CO3
Q4	Attempt any two parts of the following		
4(a)	Draw and explain the internal circuit diagram of 555-timer and explain the working of Astable multivibrator using suitable waveform and mathematical expressions	4	CO4
4(b)	With proper circuit diagram determine the output voltage of log and anti-log amplifier.	4	CO4
4(c)	With proper circuit diagram and mathematical expressions explain the following (i) Multiplier (ii) Square rooter	4	CO4
Q5	Attempt any two parts of the following		
5(a)	With the help of neat circuit diagram explain the working of a Wien's Bridge oscillator. Derive its frequency of oscillation and the gain needed by the amplifier.	4	CO5
5(b)	Draw the block diagram of the basic negative feedback topologies and enlist their properties.	4	CO5
5(c)	An amplifier has an open-loop gain of 100, an input impedance of $1k\Omega$, and an output impedance of 100Ω . A feedback network with a feedback factor of 0.99 is connected to the amplifier in a voltage series feedback mode. Calculate the new input and output impedance of the amplifier.	4	CO5